

# Griffin Halloran

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## EDUCATION

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### Indiana University

Master of Science in Computer Science

Bachelor of Science in Computer Science

Bloomington, IN

December 2020

December 2019

## EXPERIENCE

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### Autonomous Software Engineer

Feb. 2021 – Present

*L3Harris*

- Developed C++ software for autonomous surface vessels, supporting self-driving and unmanned ocean vessel operations
- Designed and implement a multi-heuristic variant of the A\* pathfinding algorithm, enabling configurable AI behaviors through interchangeable heuristic functions
- Implemented software for sensor fusion and perception systems, utilizing TCP and UDP networking to transmit and integrate sensor data across autonomous vehicle applications

### Robotic Process Automation and Machine Learning Intern

June 2020 – August 2020

*Caterpillar Inc.*

- Applied image-processing and machine learning techniques to segment and categorize handwritten check data
- Developed machine learning solutions to automatically detect, classify, and route invoices across the company
- Automated repetitive business processes by creating Robotic Process Automations (RPAs) for company-wide workflows

### Department of Navy Pathways Internship

Aug. 2017 – Aug. 2019

*Naval Sea Warfare Center, Crane*

- Developed and tested ROS-based robotic systems for the Platform Engineering Branch, working with robotic software, sensors, and autonomous system capabilities
- Helped establish a machine learning research laboratory to support on-base research and experimentation in artificial intelligence
- Created an object detection program to identify and classify objects captured from cameras

## TECHNICAL SKILLS

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**Languages:** C++, C#, Python, C, Java

**Technologies:** ROS 2, Docker, DDS, Linux, TCP/UDP, CMake, Qt

**Tools:** Mercurial, Git, Jira, Confluence, Visual Studio, Vim, LaTeX

## PROJECTS AND RESEARCH

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### Indiana University, Computer Vision Lab - Research Assistant

Jan. 2019 – Dec. 2020

- Researched architectural techniques used to design accelerators for neural networks to reduce network computational costs
- Investigate visualization methods for machine learning practices to gain a better understanding of debugging practices